Yimeng Tang

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Research Interests

- Formation and evolution of dwarf galaxies and low-surface-brightness galaxies
- Globular cluster systems as tracers of galaxy assembly
- Stellar populations and star formation histories of galaxies and star clusters
- Stellar feedback, gas flows, and environmental transformation of galaxies
- Dark-matter-deficient galaxies and galaxy–halo connection
- Emission-line diagnostics of star formation and nuclear activity

Education

Ph.D. in Astronomy, University of California, Santa Cruz — Jun 2027 (expected)
Advisors: Aaron Romanowsky, Kevin Bundy
Thesis: Decoding galaxy formation through the diversity of dwarf galaxies and their globular clusters

M.S. in Astronomy, University of California, Santa Cruz — 2024

B.S. in Astronomy (Highest Honor), University of Science and Technology of China — 2021
Advisors: Hong-Xin Zhang, Xu Kong
Thesis: Observational constraints on the evolution of nearby galaxies from IFU spectroscopy

Selected Awards & Honors

Chancellor's Dissertation Year Fellowship, UCSC — 2026
USTC Undergraduate Honorary Rank (∼ top 5%), USTC — 2021
Guo Moruo Scholarship (∼ top 2%, **highest honor for undergraduates at USTC**), USTC — 2020
National Scholarship (∼ top 2%), Ministry of Education of China — 2020 & 2018
NAOC Scholarship, National Astronomical Observatories of Chinese Academy of Sciences — 2020
Scholarship of Yan Jici Talent Program in Physics (∼ top 10%), USTC — 2020 & 2019
Outstanding Student Scholarship, Gold Prize (∼ top 5%), USTC — 2019
Outstanding Freshman Scholarship, USTC — 2017

Observing programs

As Principal Investigator:

1. Keck 2025A (0.5 night, KCWI): *Spectroscopic mapping of an ultra-diffuse galaxy with massive star-forming clumps*

As Co-Investigator with Major Contributions:

12. Roman Cycle 1 (Analysis): *Extragalactic globular clusters and galaxy formation with Roman*
11. Subaru 2026B (6 hours, HSC): *Imaging of Perseus cluster dwarf galaxies and globular clusters*
10. JWST Cycle 5 (10.3 hours, NIRCam & NIRISS): *Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances*
9. Keck 2026A (1.5 nights, KCWI): *The stellar population gradients of ultra-diffuse galaxies*
8. VLT 2025A (3 hours, MUSE): *Young and Cluster-Rich: The Progenitors of Globular Cluster-Rich Ultra-Diffuse Galaxies?*
7. Keck 2025A (2 nights, KCWI): *The stellar population gradients of ultra-diffuse galaxies*
6. VLT 2024B (17 hours, MUSE): *FCC 224: another dark matter free dwarf galaxy?*
5. Keck 2024B (1 night, KCWI): *Dark matter in NGC 1052 dwarf galaxies*

4. Keck 2024B (2 nights, KCWI): *Globular clusters and dark matter in ultra-diffuse galaxies*
3. Keck 2024B (1 night, KCWI): *A critical test for dark matter in dwarf galaxies of the NGC 1052 group*
2. Keck 2023A (1 night, DEIMOS): *Unravelling the origins of cluster ultra-diffuse galaxies*
1. Keck 2022B (3 nights, DEIMOS): *Unravelling the origins of ultra-diffuse galaxies in the Perseus cluster*

Additional Co-Investigator Programs:

3. Gemini 2026B (13 hours, GMOS): *A Benchmark Globular Cluster System for the Survey Era: GMOS Spectroscopy of NGC 1527*
2. Keck 2026B (2.5 nights, KCWI & LRIS): *Testing a Possible Second Trail of Dark-Matter-Deficient Dwarf Galaxies in the Outskirts of Fornax*
1. HST Cycle 29 (15 orbits, ACS): *A trail of dark matter-free galaxies in the NGC 1052 group*

Observing Experience: Keck Observatory (12 nights, KCWI, DEIMOS, OSIRIS), Lick Observatory (3 nights, Shane, Nickel)

Selected Talks & Presentations

- Dept. of Astronomy, Harvard University (Cambridge, Massachusetts, USA) Sep 2026
  Talk: Stellar population modeling of dwarf galaxies and globular clusters with Rubin, Roman and Euclid
- Dept. of Astronomy, Yale University (New Haven, Connecticut, USA) Sep 2026
  Talk: Puzzles of the Small: Dwarf Galaxies and Their Globular Clusters Across Environments
- 2026 Santa Cruz Galaxy Workshop (Santa Cruz, California, USA) Aug 2026
  Talk: Stellar Populations of Dwarf Galaxies in the Local Universe
  Talk: Extreme Dwarf Galaxies in the Local Universe
- 2026 Rubin Observatory Community Workshop (SLAC, Menlo Park, California, USA) Jul 2026
  Talk: A Comprehensive Study of Dwarf Galaxies in the Era of Next-generation Telescopes
  Talk: Globular Clusters in Nearby Galaxies: Numbers and Stellar Populations
- Near Field Cosmology in the Era of Big Data: Local Group and Beyond (Toronto, Canada) Jul 2026
  Talk: A Comprehensive Study of Dwarf Galaxies with Euclid, Roman and Rubin
- Kavli Institute for Astronomy and Astrophysics, Peking University (Beijing, China) Apr 2026
  Dept. of Astronomy, Tsinghua University (Beijing, China)
  Dept. of Astronomy, University of Science and Technology of China (Hefei, China)
  School of Astronomy & Space Science, Nanjing University (Nanjing, China)
  Talk: Puzzles of the Small: Diversity of Dwarf Galaxies and Their Globular Clusters Across Environments
- 10 Years of Ultra-Diffuse Galaxies: Past Perspectives and Future Prospects (Benasque, Spain) Aug 2025
  Talk: Globular Clusters as Guides: UDGs and Dwarf Galaxies in Various Environments
- 2025 Santa Cruz Galaxy Workshop (Santa Cruz, California, USA) Aug 2025
  Talk: Connection Between Dwarf Galaxies and Globular Clusters – Insights from the Perseus Cluster
  Talk: Central Emission-line Spectra of SDSS Galaxies: The Basic Evolutionary Path and its Relation to AGN Activity
- 2024 Santa Cruz Galaxy Workshop (Santa Cruz, California, USA) Aug 2024
  Talk: A New Candidate Dark Matter Deficient Galaxy with Unusual Globular Clusters
- Dwarf Galaxies, Star Clusters, and Streams in the LSST Era (Chicago, Illinois, USA) Jul 2024
  Poster: Observational Constraints on the Formation of Dark-matter Deficient Galaxies from Morphologies, Stellar Populations and Globular Clusters
- Dept. of Astronomy, University of Science and Technology of China (Hefei, China) Dec 2023
  Talk: Testing the Bullet Dwarf Collision Scenario in the NGC 1052 Group
- 2023 Santa Cruz Galaxy Workshop (Santa Cruz, California, USA) Aug 2023
  Talk: Testing the Bullet Dwarf Collision Scenario in the NGC 1052 Group

- UC Santa Cruz FLASH Talks (Santa Cruz, California, USA) Jun 2023
  Talk: Testing the Bullet Dwarf Collision Scenario in the NGC 1052 Group
- Yan Jici Physics Symposium, University of Science and Technology of China (Hefei, China) May 2021
  Talk: Observational constraints on the evolution of nearby galaxies from IFU spectroscopy
- 2019 Annual Conference of the Chinese Astronomical Society (Delingha, China) Sep 2019
  Poster: The Origin of Surface Brightness Profile Breaks of Disk Galaxies from MaNGA

Press coverage

- Scientific American: Galaxies without dark matter mystify astronomers
- Keck Observatory: Evidence of Dark Matter-Free Dwarf Galaxy Challenges Conventional Galaxy Formation Models
- Phys.org: Observations shed more light on the star formation history of galaxy NGC 2915

Service, Membership & Teaching

Member, Vera C. Rubin Observatory LSST Dark Energy Science Collaboration
Member, Roman Science Collaboration
Member, Analysis of Galaxies At The Extremes (AGATE) Collaboration
Mentor, Introduction to Research in Physics and Astrophysics (ASTR 9, UCSC) 2026 Winter & Spring
Teaching Assistant, Introduction to Scientific Computing (ASTR 119, UCSC) 2026 Winter
Teaching Assistant, Practical Programming for the Sciences (ASTR 19, UCSC) 2024 & 2022 Spring
Graduate Admissions Committee Member, UCSC 2023-2024
Prospective Student Visiting Committee Member, UCSC 2022 & 2023
Student Admissions Ambassador, USTC 2021
Teaching Assistant, Astronomical Labs (ASTR 167, USTC) 2020 Fall
Teaching Assistant, Observational Astronomy (ASTR 170, USTC) 2020 Spring

References

Aaron Romanowsky (UC Santa Cruz & San José State U) aaron.romanowsky@sjsu.edu
Pieter van Dokkum (Yale) pieter.vandokkum@yale.edu
Kevin Bundy (UC Santa Cruz) kbundy@ucsc.edu
Sandra Faber (UC Santa Cruz) faber@ucolick.org
Hong-Xin Zhang (University of Science and Technology of China) hzhang18@ustc.edu.cn
Xu Kong (University of Science and Technology of China) xkong@ustc.edu.cn

Publications

Link to my ADS publication library

First-author Publications:

9. **Y. Tang**, A. J. Romanowsky, et al., Toward Extragalactic Globular Cluster Stellar Populations: Optical+NIR SED Fitting and Initial Results in Dwarf Galaxies, 2026, to be submitted
8. **Y. Tang**, A. J. Romanowsky, N. Okabe, and K. A. Bundy, An Unusually Clumpy Ultra-Diffuse Galaxy in the Coma Cluster: A Missing Link for Globular Cluster Formation? 2026, submitted to ApJ (arXiv:2611.01653)
7. **Y. Tang**, Y. Luo, C. Zhou, The Stellar-Mass Dependence of Feedback-Driven Outflows in DESI DR1 Star-Forming Galaxies, 2026, submitted to ApJL (arXiv:2611.02580)
6. **Y. Tang**, G. S. Anand, A. J. Romanowsky, et al., New Measurements of Distances to Galaxies in the NGC 1052 Field with the Hubble and James Webb Space Telescopes: Testing the Bullet-Dwarf Origin of the Trail, 2026, ApJL, 1005, L16
5. **Y. Tang**, A. J. Romanowsky, S. Huang, et al., Connection Between Dwarf Galaxies and Globular Clusters: Insights from the Perseus Cluster Using Subaru Imaging and Keck Spectroscopy, 2026, ApJ, 998, 254

4. **Y. Tang**, A. J. Romanowsky, J. S. Gannon, et al., An Unexplained Origin for the Unusual Globular Cluster System in the Ultra-diffuse Galaxy FCC 224, 2025, ApJ, 982, 1
3. **Y. Tang**, A. J. Romanowsky, P. G. van Dokkum, et al., Testing the Bullet Dwarf Collision Scenario in the NGC 1052 Group Through Morphologies and Stellar Populations, 2025, ApJ, 978, 21
2. **Y. Tang**, B. Tao, H.-X. Zhang, et al., Unveiling the Formation of NGC 2915 with MUSE: A Counter-rotating Stellar Disk Embedded in a Disordered Gaseous Environment, 2022, A&A, 668, A179
1. **Y. Tang**, Q. Chen, H.-X. Zhang, et al., New Constraints on the Origin of Surface Brightness Profile Breaks of Disk Galaxies from MaNGA, 2020, ApJ, 897, 79

Contributing-Author Publications:

16. Z. A. Coustier, A. J. Romanowsky, **Y. Tang**, et al., Stellar Populations of Star Clusters in the Local Group Dwarf Galaxy NGC 6822 Using Keck Spectroscopy, 2026, to be submitted
15. A. Levitskiy, L. Valenzuela, D. A. Forbes, et al. (including **Y. Tang**), Diffuse Dwarf Galaxies in Galaxy Clusters: II. Stellar Kinematics, 2026, submitted
14. D. Vaz, A. Levitskiy, D. A. Forbes, et al. (including **Y. Tang**), The Rogue One: A Dark Matter-Dominated Dwarf in the Trail of Dark Matter-Deficient Galaxies, 2026, Accepted by MNRAS
13. A. Levitskiy, D. A. Forbes, J. S. Gannon, **Y. Tang**, et al., Diffuse Dwarf Galaxies in Galaxy Clusters: I. Stellar Populations and Radial Gradients, 2026, MNRAS, 551, 1568
12. M. L. Buzzo, P. G. van Dokkum, M. Hilker, et al. (including **Y. Tang**), A Twin Set of Dark Matter-Deficient Dwarfs: FCC 224 and FCC 240 as Analogues of NGC 1052-DF2 and DF4, 2026, ApJ, 1005, 15
11. D. A. Forbes, B. van Heumen, **Y. Tang**, Revisiting the distance and the globular cluster system of the remarkable galaxy UDG1 in the NGC 5846 group, 2026, MNRAS, 548, 765
10. D. Martínez-Delgado, A. J. Romanowsky, **Y. Tang**, et al., A Pearl in the Shell: an ultra-compact dwarf within the tidal debris surrounding spiral galaxy NGC 7531, 2026, A&A, 708, A172
9. M. A. Keim, P. G. van Dokkum, Z. Shen, et al. (including **Y. Tang**), Kinematic Confirmation of a Remarkable Linear Trail of Galaxies in the NGC 1052 Field, Consistent with Formation in a High-Speed Bullet Dwarf Collision, 2025, ApJ, 988, 165
8. A. J. Romanowsky, **Y. Tang**, K. A. Bundy, et al., Morphology and Stellar Populations of a Candidate Ultra-diffuse Galaxy in Early Euclid and Rubin Imaging, 2025, RNAAS, 9, 181
7. T. Li, H.-X. Zhang, W. Lyu, **Y. Tang**, et al., A Negative Stellar Mass-Gaseous Metallicity Gradient Relation of Dwarf Galaxies Modulated by Stellar Feedback, 2025, A&A, 698, A208
6. M. L. Buzzo, D. A. Forbes, A. J. Romanowsky, et al. (including **Y. Tang**), A New Class of Dark Matter-Free Dwarf Galaxies? I. Clues from FCC 224, NGC 1052-DF2 and NGC 1052-DF4, 2025, A&A, 695, A124
5. W. Lyu, H.-X. Zhang, S. Paudel, et al. (including **Y. Tang**), Dependence of Metal Enrichment of Nuclear Star Clusters on Galaxy Stellar Mass, 2025, ApJ, 979, 85
4. A. J. Romanowsky, S. S. Larsen, A. Villaume, et al. (including **Y. Tang**), Low-density star cluster formation: discovery of a young faint fuzzy on the outskirts of the low-mass spiral galaxy NGC 247, 2023, MNRAS, 518, 3164
3. G. Chen, H.-X. Zhang, X. Kong, et al. (including **Y. Tang**), Discovery of a Bimodal Environmental Distribution of Compact Ellipticals in the Local Universe, 2022, ApJL, 934, L35
2. Y. Yao, H. Liu, X. Kong, et al. (including **Y. Tang**), Subgalactic Scaling Relations with T_e-based Metallicities of Low-metallicity Regions in Galaxies: Metal-poor Gas Inflow May Have Important Effects?, 2022, ApJ, 926, 57
1. C. J. Agostino, S. Salim, S. M. Faber, et al. (including **Y. Tang**), Physical Drivers of Emission-line Diversity of SDSS Seyfert 2s and LINERs after Removal of Contributions from Star Formation, 2021, ApJ, 922, 156